

Thread - 2026-04-19

<https://x.com/RiikonenMarko/status/2045798300348035307>

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13/14 Dec 2025, Rovaniemi. A display in Anttilanvaara. I wonder about the faint feature marked by arrows. Integrated flip-sun blue edge from snow reflection? If so, then it is curious that it appears on the rear-hemisphere where the ground was not lit.

Another possibility would be (counter)zenith arc from low situated surrounding lights, but all lights are far away and in the direction opposite to the lamp there was nothing. For reference of this type thing, see the lunar display on the night of 13/14 December 2013 in Koli, Finland, that included both the counterzenith arc and its Dufay from the ski center floods:

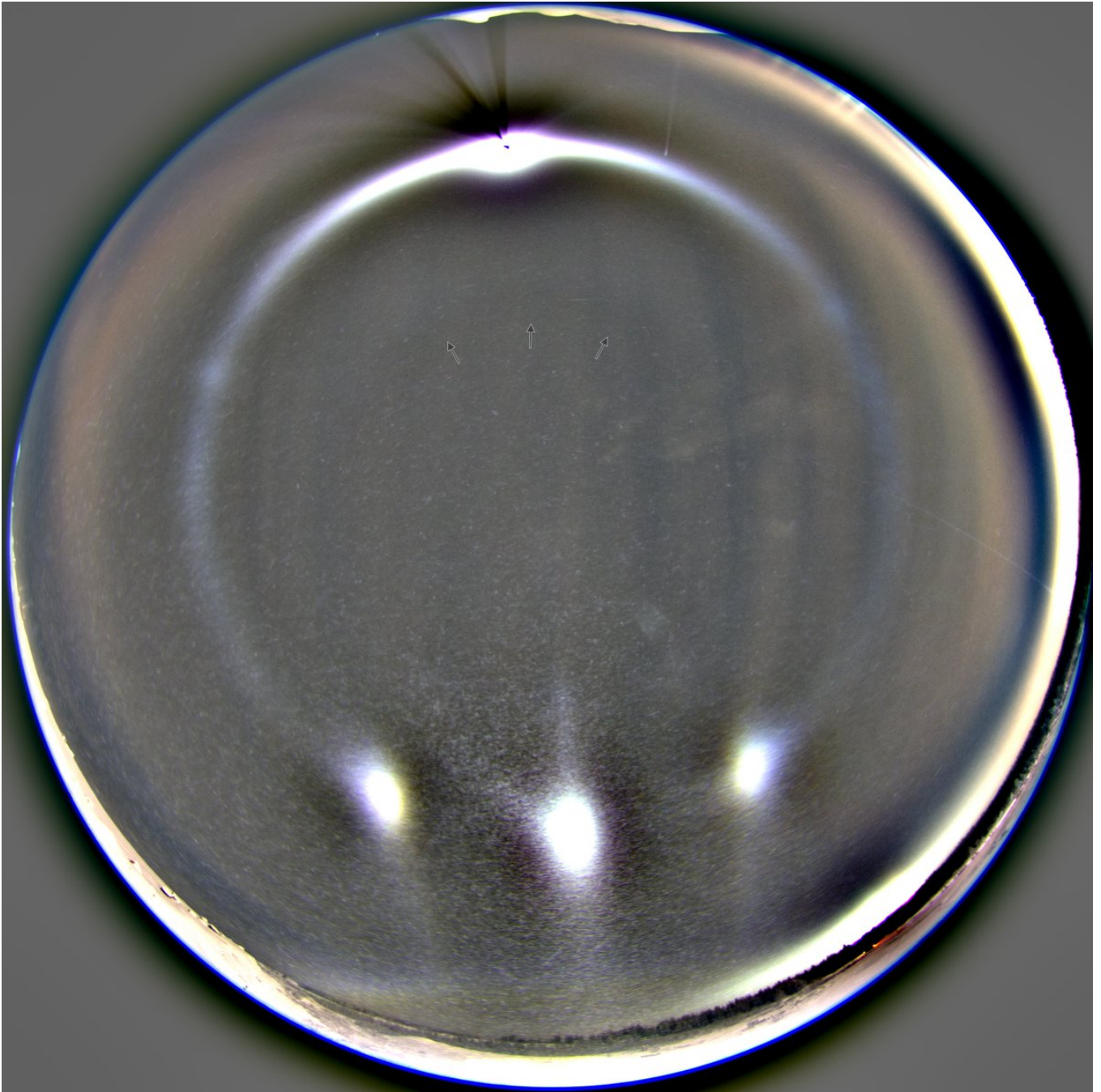
<https://t.co/i2XqMRp7IP>

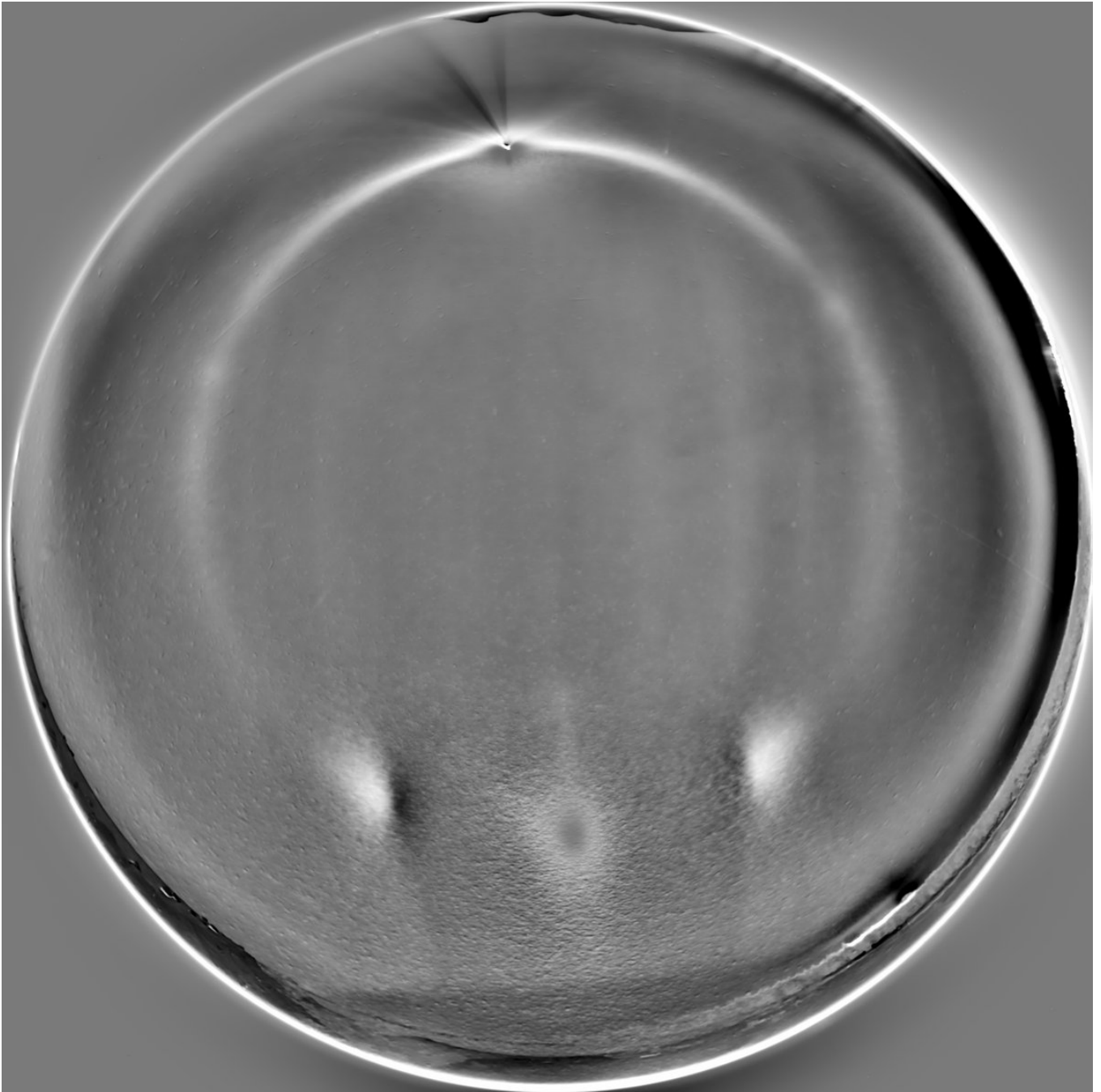
<https://t.co/pslypmy8AQ>

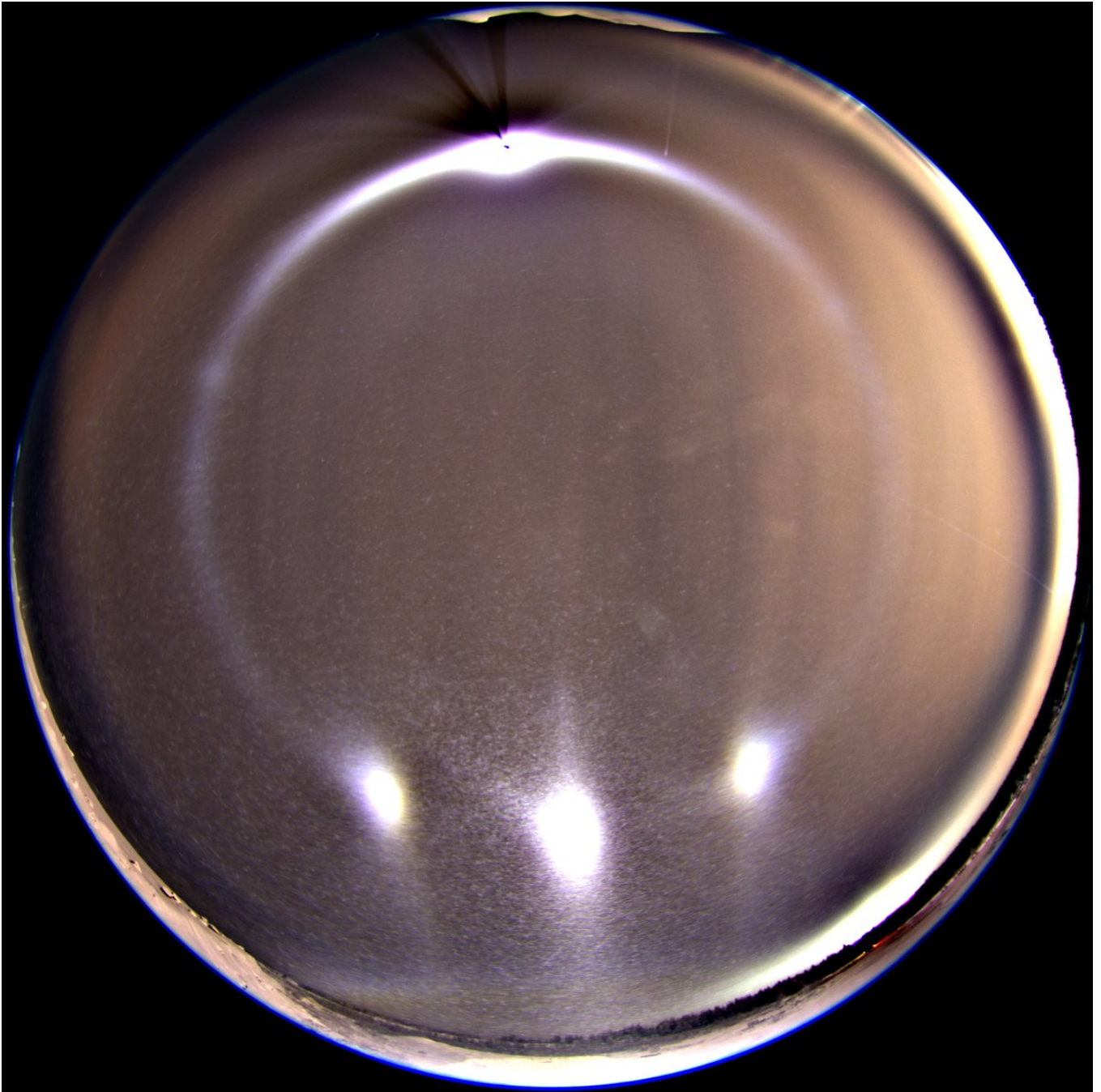
Well, maybe this faint effect is not worth all the ink. Yet I was not able to rest until next evening I revisited the site and reconstructed the situation, as shown in the next post.

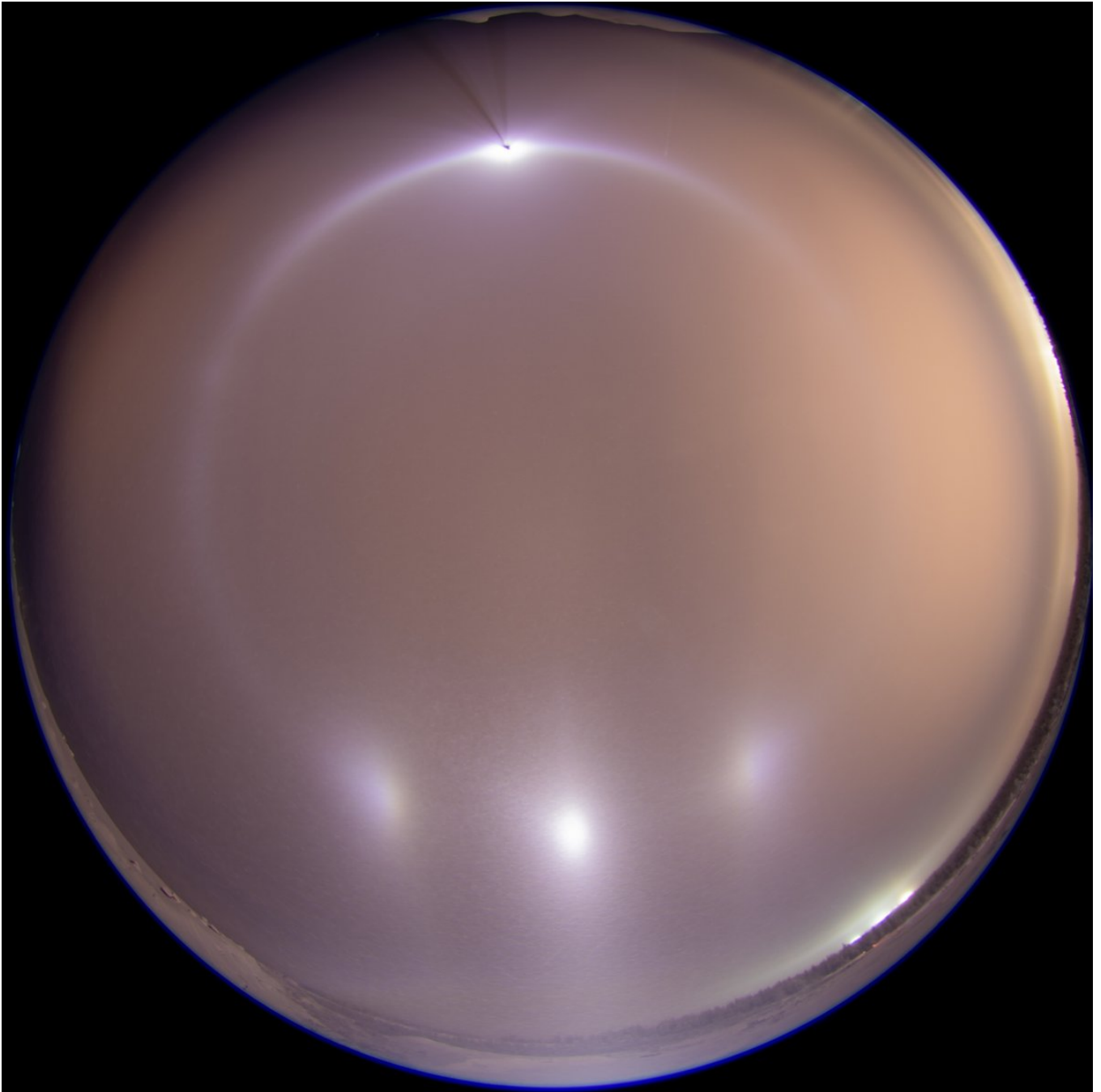
Another thing to note is the countersun spike to the upper left. This must be an aerodynamic anomaly, airflow accommodating to an obstacle and crystals tilting along for the ride. Is the camera, the steep slope below, or even the nearby mound on the upwind side the culprit? Wind was from the right.

17 min exp, 34 x 30s. Hard to say about the lamp elevation. But countersun is way above 46° halo. Maybe around -40 degs.







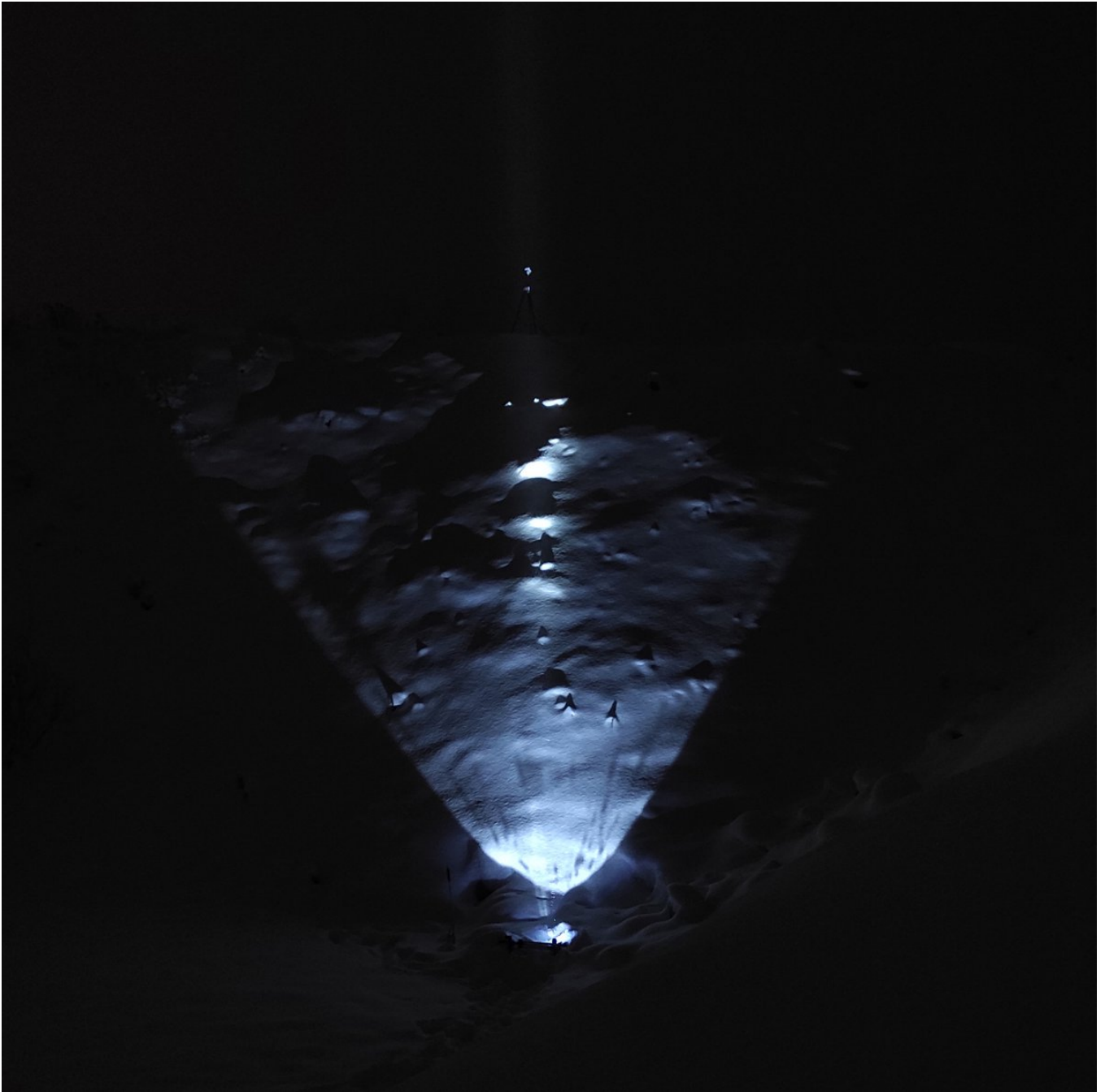


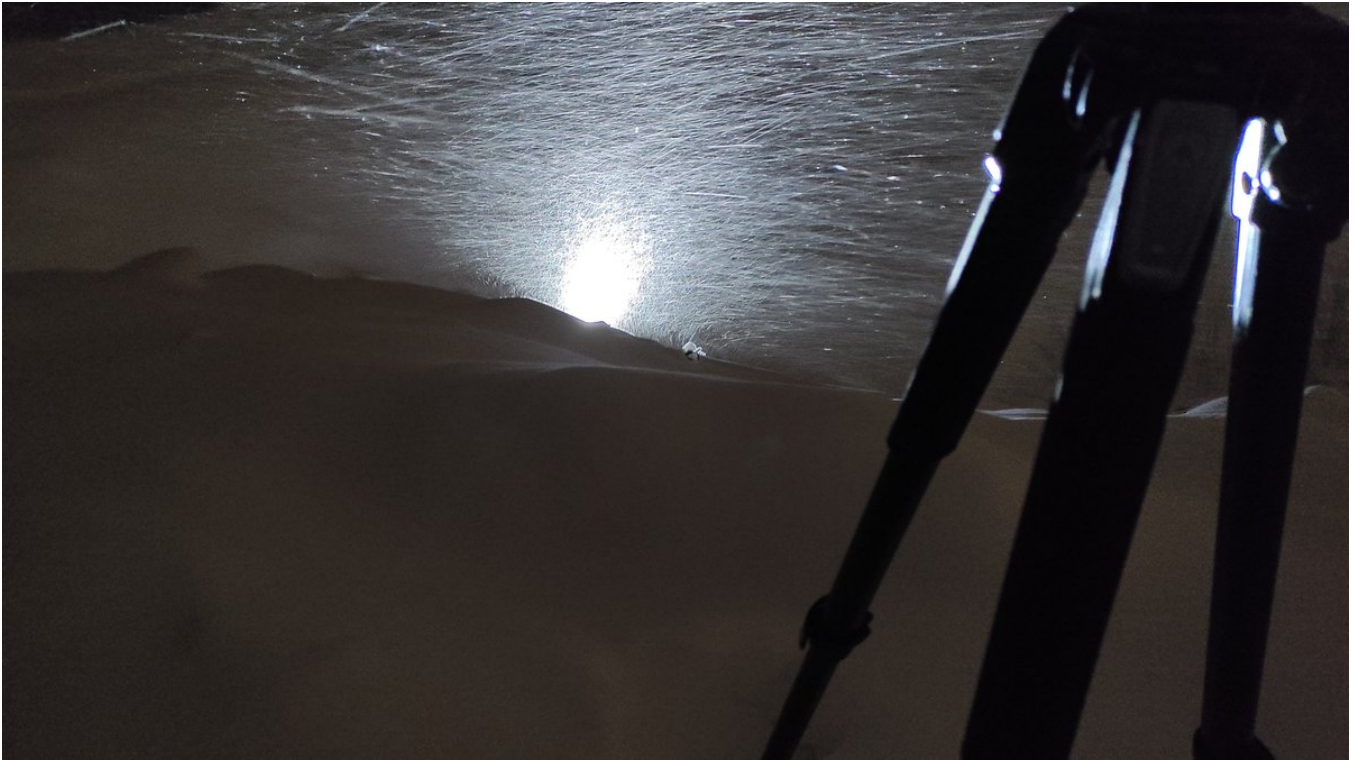
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13/14 Dec 2025, Rovaniemi. Reconstruction of the above post's situation. It was possible to do pretty accurately as the tripod leg holes in the snow were still visible despite it having snowed and drifted.

Only the slope was lit by the Javelot Turbo. Up where the camera was, the ground was in shadow (third photo).

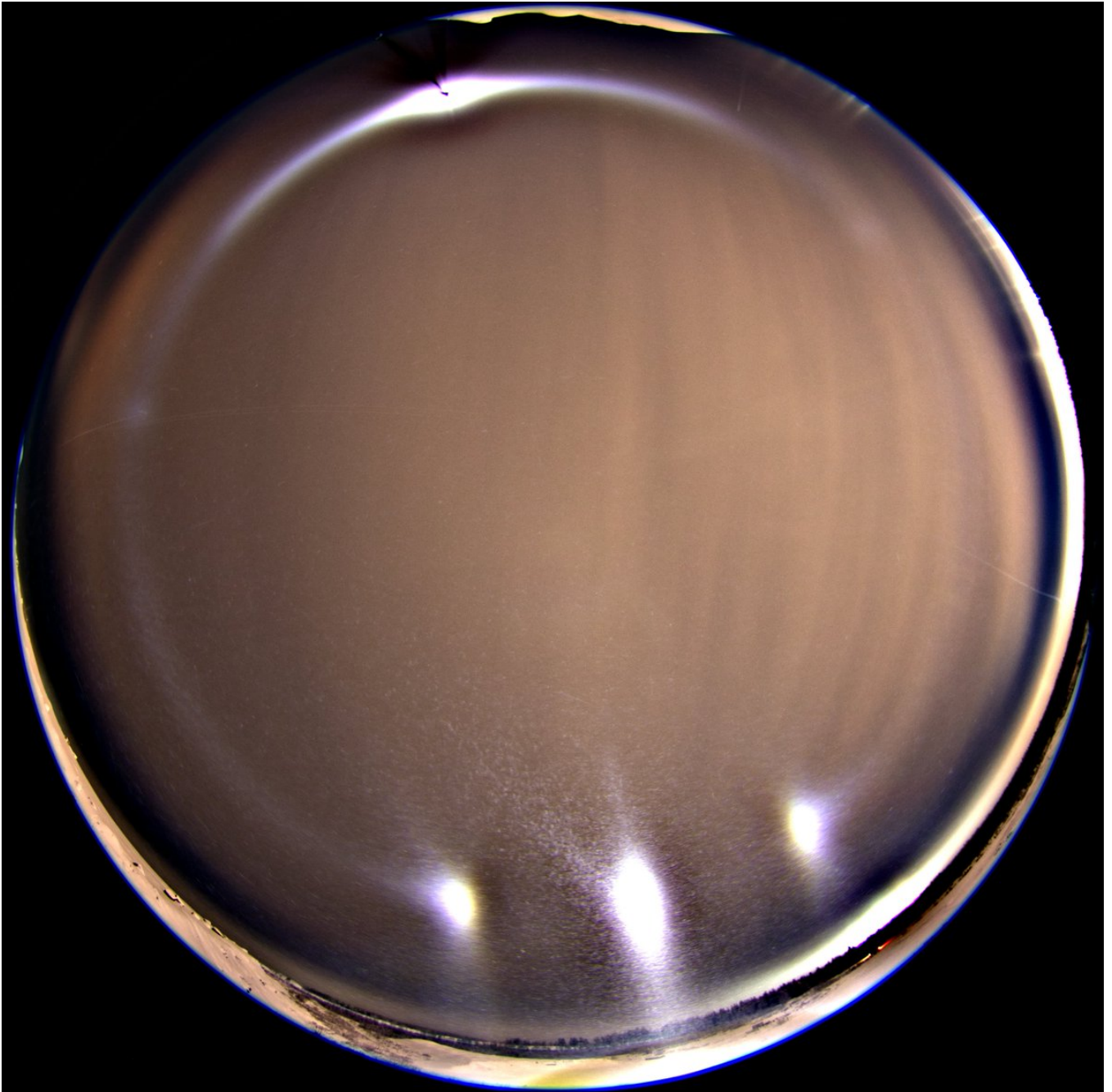
How about reflections from the tripod, blocker and camera to explain the rear-hemisphere arc?



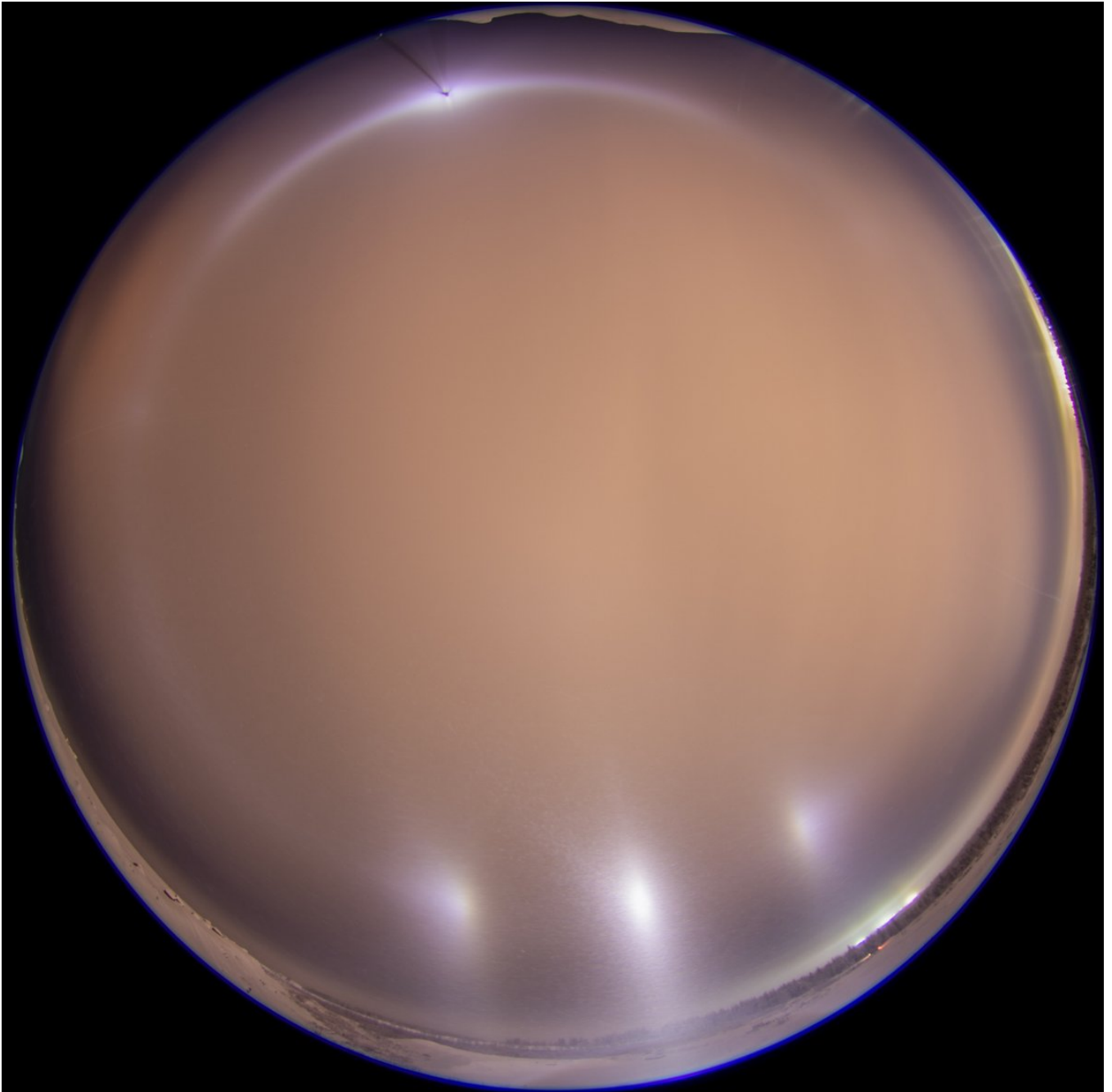


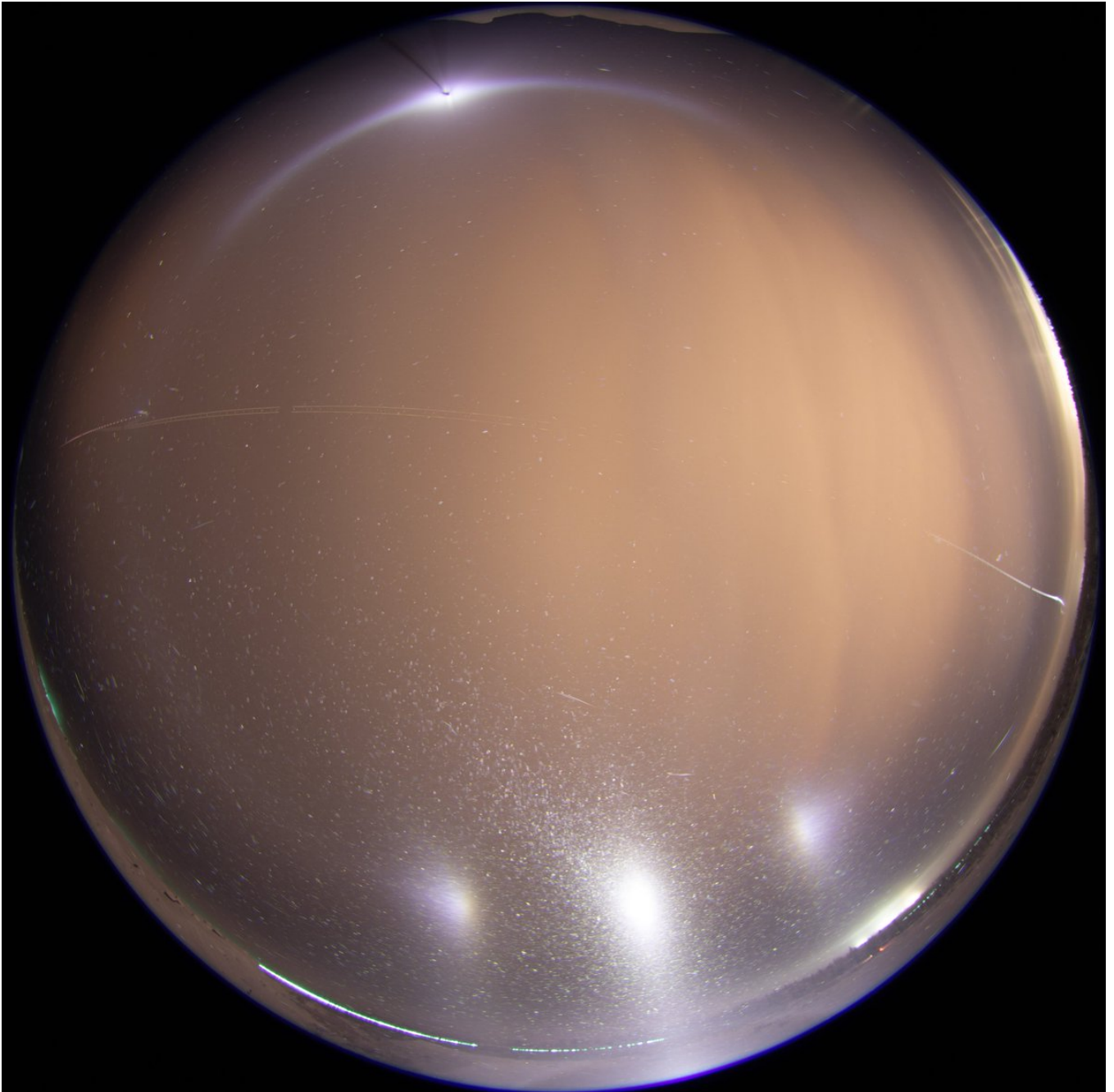
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13/14 Dec 2025, Rovaniemi. This is nearer-horizon light stack, also 17 min exposure. No arc is seen in the back-hemisphere, the only anomaly is the spike from flipsun. A short diffraction pillar aka Mikkilä's Soul is visible.



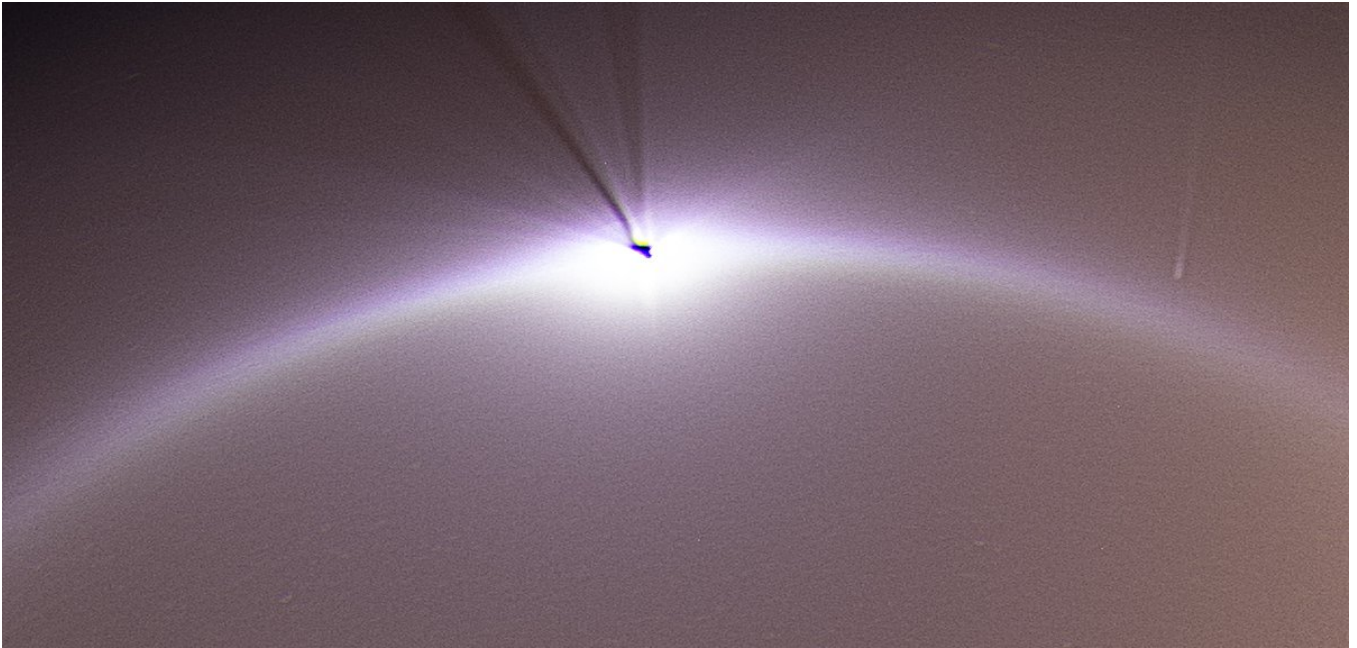






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13/14 Dec 2025, Rovaniemi. This close-up showing the Soul is from the first set. Is this the furthest elevation observation? In regular hexagonal plates, shouldn't the pillar transform into a circular glory as the light nears poles?



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13/14 Dec 2025, Rovaniemi. Earlier on, I had written a pdf to desktop drawer of this display. This is what I wrote about the next stage:

Then I tested if the integrated divergent flipsun blue edge can be summoned in a positive-light set-up, camera on the ground so the snow around is brightly lit by the hot spot (10 x 30s). No success. The only thing I can associate with the lit snow is the zenith spot, conforming to the elongated shape of the lit area.

But the display was already in a decline, so I think this needs to be replicated in a stronger display, and preferably in a clear-sky one. And also take photos so that the lamp is right under the tripod, pointing down.

Indeed, later in February I happened on a stronger display which contained the integrated flipsun blue edge and also managed to pull off the lamp-right-under-the tripod experiment, as recounted in this thread: <https://t.co/fPLWOHVUgo>

I am not sure, though, if the spot at the zenith (in both displays) should be regarded as zenith spot or just countersun from the elongated lit patch on the ground. Maybe the distinction is unnecessary. Far as I can tell, this is not zenith spot from direct lamp light. In the February display the spot at the zenith is extending a tail in the direction opposite of the lamp <https://t.co/3whCohStCK> If it was formed directly from the lamp photons, the tail would be towards the lamp.

In the original pdf there is some more info, like simulation attempts on those parhelion related arcs: <https://t.co/7IHs7s58E3>

My take on some things have changed in the intervening time, so not everything is in line in the pdf with what I have written in this thread.

